Recap: Defined $\mathbb{R}$ → Completeness axiom

Characteristics of $\mathbb{R}$ → Archimedean property

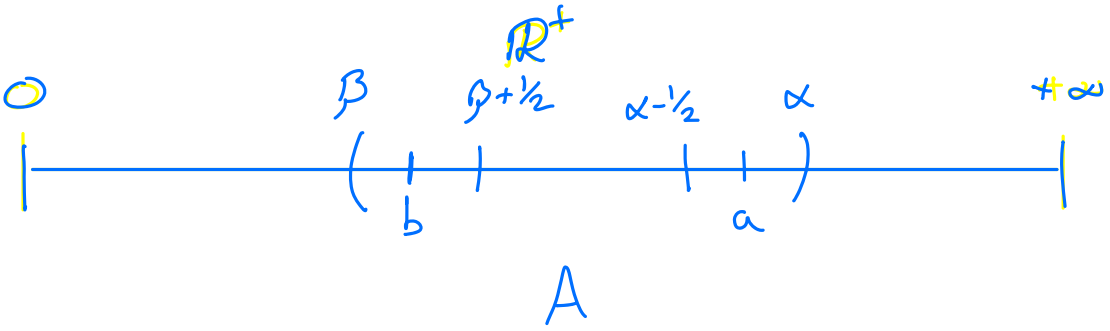
→ Nested Interval property

→ Density of $\mathbb{Q}$ in $\mathbb{R}$.

Theorem 4.2: Suprema and Infima Analytically

i) Let $\alpha \in \mathbb{R}$ be an upper bound for a non-empty set $A \subset \mathbb{R}$. Then $\alpha = \sup(A)$ iff for every $\varepsilon > 0$ there exists an $a \in A$ such that $\alpha - \varepsilon < a$.

Pic:



(ii) Let $\beta \in \mathbb{R}$ be a lower bound of a non-empty set $B \subset \mathbb{R}$. Then $\beta = \inf(B)$ iff for every $\varepsilon > 0$, there exists $b \in B$ such that

$$b < \beta + \varepsilon .$$

Proof: (i) ($\Rightarrow$) Assume $\alpha = \sup(A)$. Then α is the least upper bound of A. Hence, $\alpha - \varepsilon$ is not an

upper bound for A for every $\varepsilon > 0$. That is, there exists $a \in A$ such that $\alpha - \varepsilon < a$.

($\Leftarrow$) Now assume $\alpha \in \mathbb{R}$ is an upper bound for A and for every $\varepsilon > 0$, there exists $a \in A$ such that $\alpha - \varepsilon < a$. This means any value less that α is not an upper bound for A. Thus any other upper bound, say $u \in \mathbb{R}$ for A, must satisfy $\alpha \leq u$.

Hence, $\alpha = \sup(A)$. ☑

(ii) Exercise

Example 4.3: Infima and Suprema

(i) Let $A = \left\{ 1 - \frac{1}{n} : n \in \mathbb{N} \right\}$. Then $\sup(A) = 1$.

Proof: Since $0 \in A$, A is non-empty. As $n > 0$ for all $n \in \mathbb{N}$ it follows that $1 - \frac{1}{n} < 1$ for all $n \in \mathbb{N}$. That is, 1 is an upper bound for A. [We need to find a value $N \in \mathbb{N}$ such that

$$1 - \varepsilon < 1 - \frac{1}{N},$$

for all $\varepsilon > 0$. Rearranging gives

$$-\varepsilon < -\tfrac{1}{N}$$

$$\Rightarrow \varepsilon > \tfrac{1}{N}$$

$$\Rightarrow N > \tfrac{1}{\varepsilon}.$$

We need to then choose a value of $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. By the Archimedean property, we know such a natural number always exists.

Given $\varepsilon > 0$, by the Archimedean property there exists $N \in \mathbb{N}$ s.t. $N > \frac{1}{\varepsilon}$. Hence,

$$1 - \varepsilon < 1 - \tfrac{1}{N}.$$

That is, for every $\varepsilon > 0$, there exists an $a \in A$ s.t. $1 - \varepsilon < a$. Hence, $\sup(A) = 1$. [2]

(ii) Let $B = \{n - 1 : n \in \mathbb{N}\} = \{0, 1, 2, 3, \ldots\}$.

Then $\inf(B) = 0$.

Proof: Clearly B is non-empty. Since $n \geq 1$ $\forall n \in \mathbb{N}$ it follows that $n-1 \geq 0$ $\forall n \in \mathbb{N}$. That is 0 is a lower bound for B. We need to find an $N \in \mathbb{N}$ s.t

$$N - 1 < 0 + \varepsilon$$

for all $\varepsilon > 0$. Clearly $N = 1$ satisfies this inequality for all $\varepsilon > 0$. Hence $0 = \inf(B)$. ∎

Defn 4.4: Max and Min of a Set

Let $A \subset \mathbb{R}$. Then if there exists an $M \in A$ s.t $a \leq M$ for all $a \in A$ the set A is said to have a maximum, denoted by $\max(A)$. Similarly, if there exists $m \in A$ such that $a \geq m$ for all $a \in A$, the set A is said to have a minimum, denoted $\min(A)$.

Example 4.5: $\sqrt{2}$ is Real Number

Consider the set

$$A = \{ x \in \mathbb{R} : x^2 < 2 \text{ and } x > 0 \}.$$

By the completeness axiom, $\sup(A)$ exists. Recall in $\mathbb{Q}$ this was not the case. For this proof you would show that $\sup(A) = \sqrt{2}$. Proof outline is to show that $\sup(A)$ is a number such that its square is 2.